

Jonathan Liu

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EDUCATION

University of California: Santa Barbara

Bachelor's of Science, Computer Engineering

Santa Barbara, CA

Sep 2026 — Jun 2030

Del Norte High School

High School Diploma, Certification in Software and Systems Development / Engineering Design

San Diego, CA

Aug 2022 — Jun 2026

- AP Scholar with Distinction Award
- National Recognition Program - School Recognition Award for Outstanding Academic Achievement

WORK EXPERIENCE

FTC Robotics Programming

Mar 2025 — Jun 2026

22105 Runtime Terror

San Diego, CA

- Developed computer vision pipelines for course correction during autonomous pathing and AprilTag relocation
- Programming PTO clutch based hanging mechanism to suspend the robot in the air
- Implementing software patches at FIRST World Championship
 - Inspire 3 Award / Thompson division finalist alliancat 2025 Michiana Premier Event
 - 1st place alliance / Inspire 1 in 2026 San Diego Gauss League Championships
 - Inspire 2 at 2026 San Diego Regional

Phylogenetics Data Analyst

May 2025 — Aug 2025

UCLA Pellegrini Lab

(remote) San Diego, CA

- Computed and analyzed differences amongst .newick trees with Python
- Constructed phylogenetic trees comparing UCLA's UMAP database's Hypocreales taxonomy to NCBI

Surgery System Research Intern

Jan 2025 — Jun 2025

UC San Diego Institute of Engineering in Medicine

San Diego, CA

- Implemented an ArUco C# based AR tool as an X-Ray free alternative to fluoroscopy spine imaging
- Compiled findings into abstract and poster, presented at ICBEB 2025 winning Best Poster Award
 - Development of an Augmented Reality Surgical Navigation System Based on Multiple ArUco Markers

CS Education Research Intern

Oct 2024 — May 2025

UC San Diego

San Diego, CA

- Scraped 35,000+ Scratch projects via Python web-scraper for program code in .json format; saved in database with SQLite
- Quantified trends in program code based on "opmode" block frequency in project code
- Presented abstract and findings in poster format in a program symposium
 - Insight On K-12 Computer Science Education Trends Through Scratch Project Analysis

Library Volunteer

Jun 2024 — Jan 2026

San Diego County Library

San Diego, CA

- Provided customer service, shelving books, and community events, e.g., robotics scrimages, chess tournaments

FRC Robotics Programming

Nov 2022 — Jul 2025

Team Optix 3749

San Diego, CA

- Optimized teleoperation through developing "On The Fly" dynamic routing and driving to precise positions
 - Innovation In Control Award at San Diego Regional

PROJECTS

Relay (github.com/Superposition-Development/Relay)

Aug 2025 — Present

- Maintaining based full-stack Go terminal application for real time communication alongside WebRTC based calling
 - Operates off of decentralized HTTP servers and SQL database; open source to give users privacy by routing without a third party

SKILLS

- **Languages:** HTML/CSS, JavaScript, Java, C#, Go, Python, C++, MatLab, SQL, Bash, Lua
- **Technologies:** Git, OpenCV, CUDA, Linux, Arduino, CAD, Blender, Photoshop, Databases